

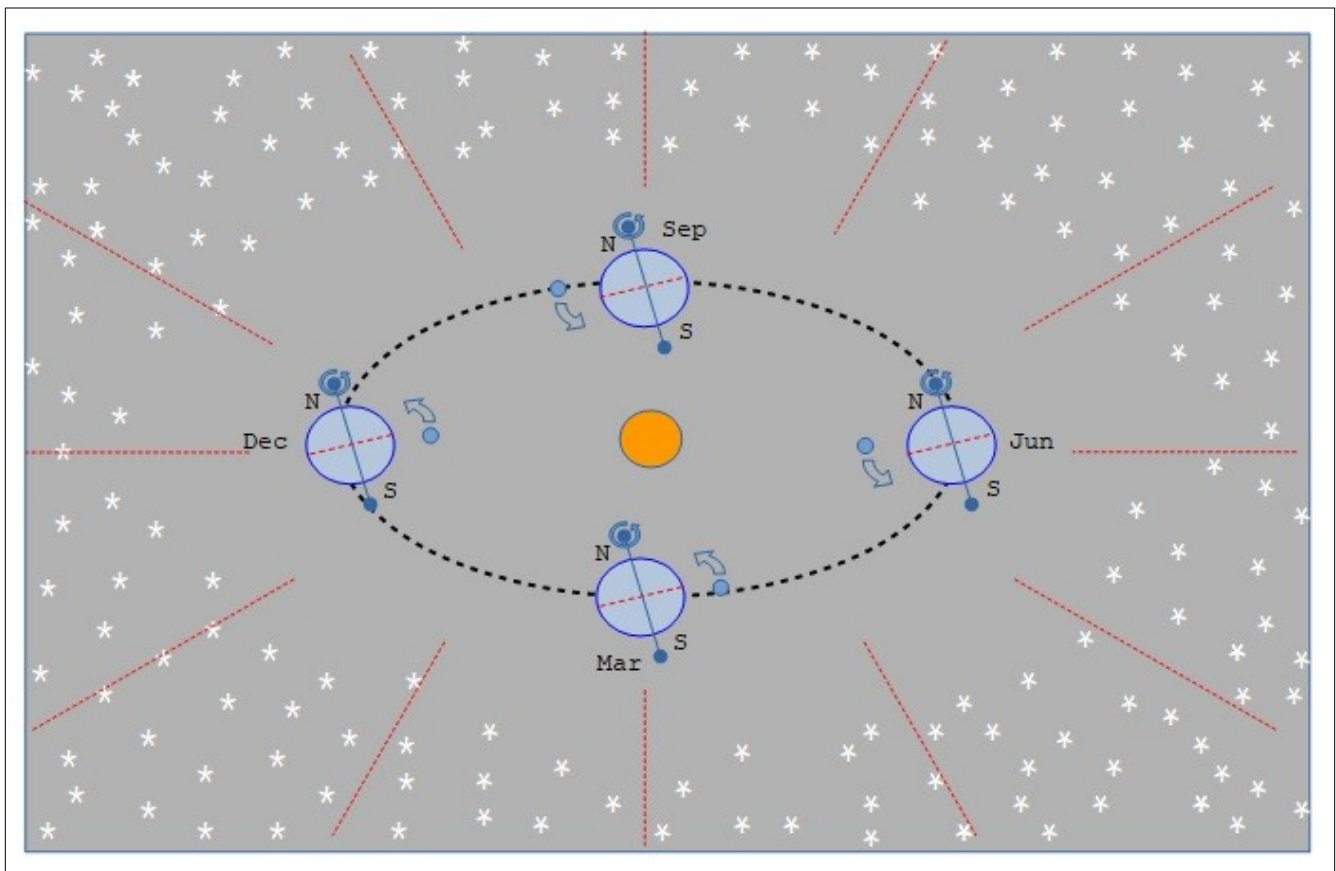
Calendars of Hindus

A calendar in general is about time keeping, accounting for and organizing the time line. In a broader sense it perhaps started as need for food gathering and cultivating for any culture, because in a given span of time the seasons and weather changes in a periodic manner – synchronizing with nature and its events. The periodic changes is based on the fact that Earth revolves around the Sun. We call one full revolution around the Sun as 'year'.

Movement:

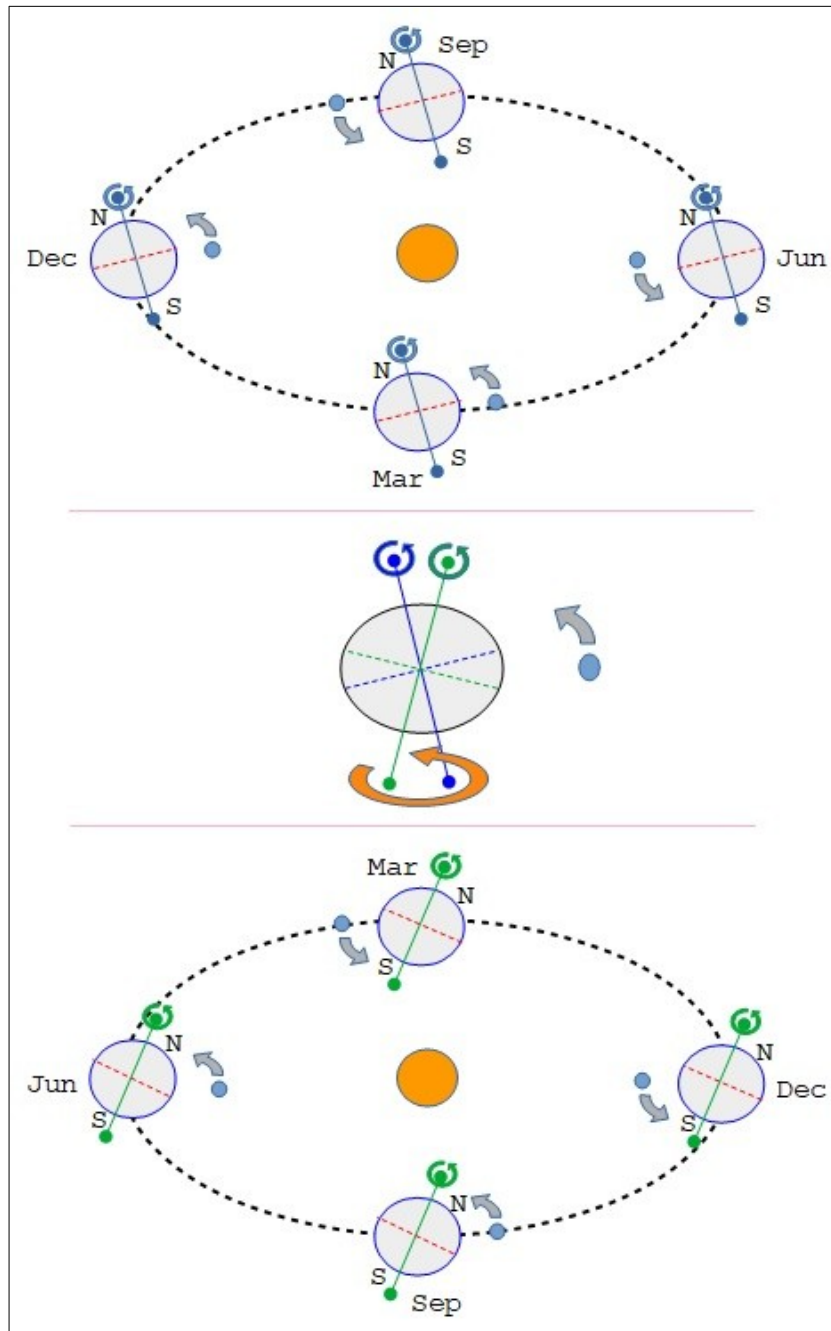
So keeping a time, requires us to first understand everything that moves – real and apparent – with respect to where we live called Earth. Looking at the following picture (Picture – 1) we can see that following things move –

- a) *Earth spins around its axis:* About little less than 24 hours (1 day)
- b) *Earth revolves around the Sun:* About little over 365 days
- c) *Moon revolves around the Earth:* Little over 27 earth days
- d) *Moon spins around its axis:* Almost the same time as revolving around earth – giving an apparent impression of not spinning
- e) *The Star that we see at night also change periodically:* Apparent motion due to above revolution of earth around the sun



Picture – 1: Movements with Respect To Earth

In addition to this there is another motion that earth has – that is called ‘precession’. To understand ‘precession’, imagine a toy ‘Spinning Top’. It ‘spins’ around the axis, but the axis also ‘wobbles’ around in circles when observed from top. That circular wobbling of the spinning axis is called ‘precession’. Likewise, the axis of earth which is at a slight inclination with respect to the Earth-Sun plane also wobbles around the axis – that is the ‘precession’ motion of earth. The following picture (Picture-2) illustrate how the precession motion of Earth changes the seasons with respect to the relative position of Earth and Sun, as observed from a distant location. The axial precession takes about 26000 Earth years to complete one precession cycle.



Picture – 2: Precession of Earth

Reference Points:

Setting aside the 'accuracy' of the values itself, the 'durations' of the movements mentioned above also differ based on the 'reference point' we choose. We can generally acknowledge 2 reference points in this context that is immediately apparent to us.

Sidereal – This comes from the Latin root 'sidereus' meaning 'belonging to star'. This is as if observing the movements from a distant location as the 'reference point', and hence can be considered as being 'absolute' and irrespective of what we see from earth (like a 'bird's eye view').

Synodic – This comes from Latin 'synodus' as in 'meeting' - similar to 'syn' like in 'synchronizing'. This is as if observing the movements as we see it from Earth as the 'reference point', and hence can be considered as being 'relative' to our location.

For example, if the Earth wasn't spinning at all but is revolving around the Sun – then after one full revolution, we have zero sidereal day but '-1' (lost a day) Synodic day. That's because as observed from a distant star, earth did not spin at all and so has zero Sidereal day. However, for the person living on earth, would observe that the Sun apparently rises from the west and sets on the East once in that year – that's one full day, but in opposite direction – that is one Synodic day, (rather its '-1').

So when we say there are about 365 days in a year – those are 'Synodic days'. It means that we see the Sun rise and set 365 times in a year. The 'Sidereal days' are then about 366 days (one more than Synodic) – because as seen from a distant star, the Earth actually spun 366 times in that year.

The Year:

We know that the Earth spins about its axis and revolves around the sun. We also know that the axis of spinning is tilted with respect to the Earth-Sun plane (Picture – 1 Above). So we have changing seasons through the year. One complete revolution around the Sun with periodic and repetitive seasons makes one year – we call that 'Synodic' year as 'Tropical' – based on the Greek root 'tropicos' meaning 'turn or change or solstice'.

Looking at Picture-2 above, we see that since earth's axis wobbles, called precession, the absolute 'location' of a particular 'season' changes with time. Considering that the axial pecession cycle is about 26000 years, we can see that in about half that time (13000 years), the earth's location for the same season is on the opposite side of sun. So with the known retrograde direction of wobble (precession), looking from a distant star, the 'Tropical Year' makes for *slightly less than 360 degree revolution* around the sun. The full 360 degree revolution around the Sun is then called 'Sidereal' year.

The angular difference between the 'Synodic/Tropical Year' (shorter) and the 'Sidereal Year' (longer) is about $360/26000$ degrees [= ~ 0.014 deg = ~ 50 arc seconds] (about 20 minutes per year). That means, after one full Tropical year, we on Earth see the exact same apparent/relative location of Sun with respect to Earth (experiencing same 'season') – but it takes 20 more minutes for Earth to get to the same exact absolute location in space with respect to Sun, as seen from a distant star.

The wobbling of Earth about its axis (precession) in Sanskrit is called 'ayana'. The word 'sa-ayana' is used when the precession is taken into account, while the word 'nir-ayana' is when the precession is

not taken into account. So Synodic/Tropical in Sanskrit is 'saayana' system and the Sidereal is 'nirayana' system.

Calendars:

So now the question is how do we measure the 'Year' - that is a Calendar!

Synodic/Tropical Solar Calendar:

Some parts of India such as sections of Tamilnadu, Kerala etc, take the 'Synodic/Tropical Year' (saayana) and divide that by the number of rotations Earth has in that duration – which is about little over 365 days. These days are further divided into variable size months with fixed size weeks. The festivals, celebrations, events etc of the year are fixed on this Calendar. Please note that the 'Sidereal' rotations of Earth has no immediate practical value in the Calendars. So the mention of Earth's rotations must always be assumed to be 'Synodic' type unless mentioned otherwise.

This Calendar is technically closest to what is used in the west called 'Gregorian Calendar', which is derived from after corrections to 'Julian Calendar'. While the western calendar is the most widely used one in the world, it is derived on the same principles as the ancient Solar Calendar of India.

Sidereal Solar Calendar:

A Sidereal (nirayana) variant of Solar Calendar, the one which is independent of precession, does exist in some parts of India. It is however not of much broader value. That is because it has no immediate practical application to the seasons and related events such as agriculture etc.

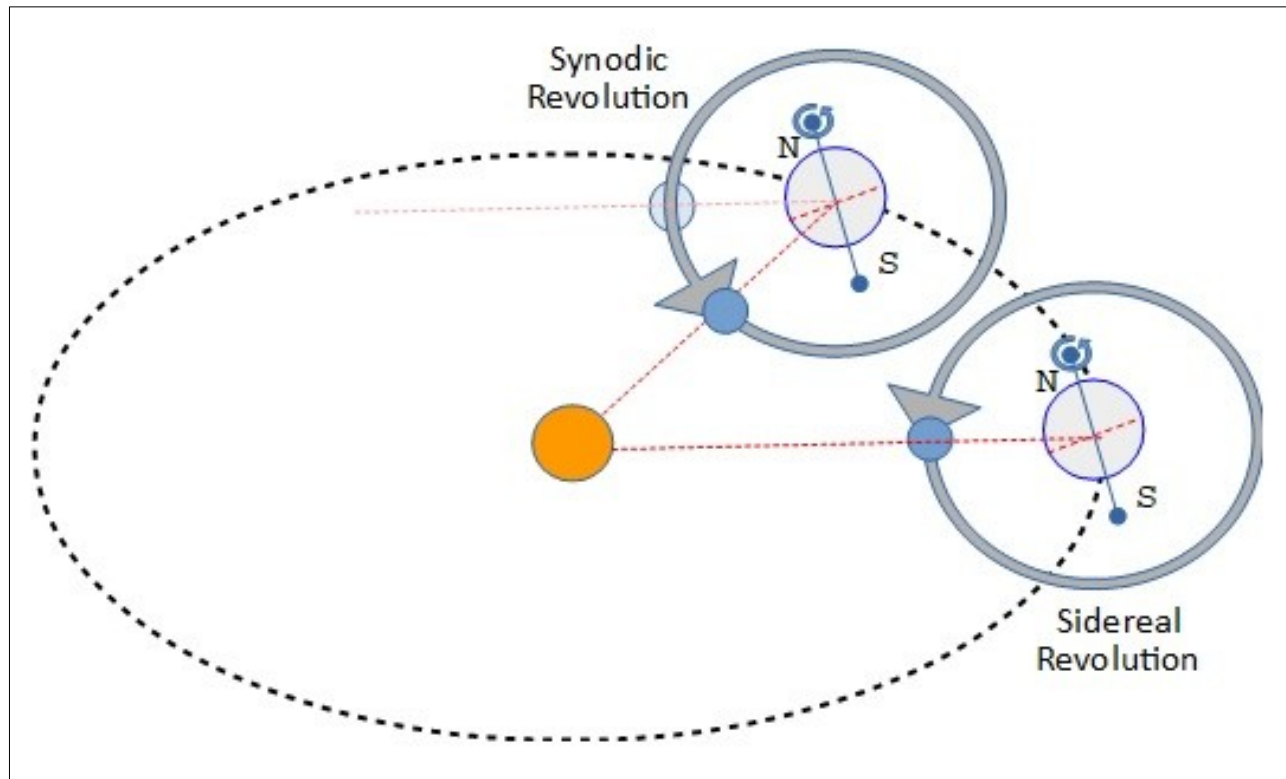
However, a more complex variant of the nirayana and saayana system combined, called 'panchanga', is very ancient and very widely used in India for many other purposes. The details of that Calendar is described below.

Synodic/Tropical Luni-Solar Calendar:

This type of Calendar is more widely used across the country compared to the pure 'Solar' variant mentioned above. In this calendar, the days of the 'Synodic/Tropical Year' (saayana) is still based on the rotation of Earth. But the duration of a month is fixed and is roughly based on the number of 'Earth days' moon takes to revolve around the earth (Synodic revolution of moon) – which is about little more than 29.5 days, giving 12 full months in a year and some extra days. Please note that moon takes little over 27 days for a Sidereal revolution around earth – which is absolute one complete revolution as observed from distant Star (Picture-3). The Synodic revolution of moon is – the moon position getting back to being at the 'same location', relative to observation made from Earth. This is longer because Earth is moving around Sun and makes an angular progress in those days.

In this Synodic revolution, the moon completes an apparent 360 degrees around earth. So the 29.5 days is divided to 30 equal units called '*tithi*' (which is roughly like 'days'). Each tithi is 12 degree (360/30) angular change of moon in the sky. More details in 'panchanga' section below.

Dividing the 'Synodic/Tropical Year' into equal size months leaves several days (about 10 days) that are unaccounted for. So roughly about every 32 months, an extra month is added to the year called 'adhika maasam'.



Picture – 3: Moon's Synodic and Sidereal Revolution Around Earth

Lunar Calendar:

A pure 'Lunar Calendar' that is based exclusively on the number of days it takes for the Moon to revolve around Sun (about 30 days), has been done before in ancient times. It is kind of a 'preliminary' step to the aforementioned more sophisticated Luni-Solar Calendar – which also accounts for the revolution of Earth around Sun. Such a pure Lunar Calendar is rather rudimentary and has not much practical value in most societies – as it has no notion of 'seasons' which is essential for many important things such as agriculture. This type of Calendar is still used in Islamic societies.

Panchanga (Sidereal+Synodic):

Panchanga literally translates to 'five limbs' (pancha + anga). It is a Sidereal-Synodic based complex Calendar. The five 'limbs' are tithi, vaara, nakshatra, yoga and kaarana. Please note that these 5 anga (limbs) are very intrinsically dependent on several other concepts. The following is a description of each from the dhaatu ('root of root' or 'concept/idea' represented with √) of each word.

tithi:

√ tij – be precise, sharp, accurate etc. It's the concept of 'getting it exact'

From it came the derived words such as –

atithi – a guest, one who has no ‘precise’ schedule/time of arrival

tithi – From this dhaatu *vtij* came the word ‘tithi’ which means ‘a precise division’ and is roughly translated in English in the context of astronomy and astrology as ‘exact division of lunation’.

The revolution of Moon around the Earth takes about 29.5 days (Synodic). The revolution is also by definition 360 degrees, as observed from Earth (Synodic). The tithi is a way of dividing the arbitrary 29.5 Earth days into precisely 30 equal units. The tithi is then the quasi-equivalent to a ‘day’ for the time keeping purposes.

Since one complete Synodic rotation of Earth about its axis (divasa) is also precisely divided into 24 hours (hora), this superimposition of tithi (30 precise units in a Lunar month, 29.5 Earth days) over Earth days (24 exact units in a day, from sunrise to sunrise) also intrinsically gives two vectors of time keeping for a ‘day’. The two vectors being – 24 hour (hora) Earth day (divasa) and the 30 units of a Lunar month (maasa) called tithi.

The 30 tithi are:

sukla/suddha paksha (waxing moon side)	Tithi name	krishna/bahula paksha (waning moon side)	Tithi name
1 st tithi [1]	pratipada/padyami	1 st tithi [16]	pratipada/padyami
2 nd tithi [2]	dwitiya/vidiya	2 nd tithi [17]	dwitiya/vidiya
3 rd tithi [3]	tritiya/tadiya	3 rd tithi [18]	tritiya/tadiya
4 th tithi [4]	chaturthi/chaviti	4 th tithi [19]	chaturthi/chaviti
5 th tithi [5]	panchami	5 th tithi [20]	panchami
6 th tithi [6]	shashti	6 th tithi [21]	shashti
7 th tithi [7]	saptami	7 th tithi [22]	saptami
8 th tithi [8]	ashtami	8 th tithi [23]	ashtami
9 th tithi [9]	navami	9 th tithi [24]	navami
10 th tithi [10]	dashami	10 th tithi [25]	dashami
11 th tithi [11]	ekadasi	11 th tithi [26]	ekadasi
12 th tithi [12]	dwadasi	12 th tithi [27]	dwadasi
13 th tithi [13]	trayodasi	13 th tithi [28]	trayodasi
14 th tithi [14]	chaturdasi	14 th tithi [29]	chaturdasi
15 th tithi [15]	pournami/ poornima	15 th tithi [30]	amavasya

In a Synodic/Tropical year, the Moon completes 12 full Synodic revolutions around Earth – leaving around 10 extra days in the year. That brings us to the next topic of dividing the Year into Months (maasa).

raasi:

vrāś – a sum or collection of something

rāṣṭrā – from *vrāś* and *vtrai*

vtrai = (dhaatu) protector and defender

rāṣṭrā = Country – a protected and defended collection of land

raasi – From this dhaatu *vrāś* came the word ‘raasi’ which means ‘a collection of or set’ and is roughly translated in English in the context of astronomy and astrology as ‘zodiac’.

Having understood and studied the movement and position of Moon with respect to Earth, it is clear that it’s a repetitive event occurring every ‘lunar month’ (maasa). The relatively ‘stagnant’ sky in the background with stars, which forms a distant steady reference point, then functions as a guide to understanding the relative position of Earth-Moon system with respect to Sun. In that context, we know that there are 12 complete ‘Lunar months’ in a year (and some 10 extra days). This then forms the basis of dividing the 360 degree sky into 12 units called ‘raasi’ and identifying a distinct ‘collection or clusters of stars’ in each unit. The raasi then works as the ‘celestial scale’ of 12 units in 360 degrees, through which our Earth is navigating.

The 12 raasi are:

raasi name– samskrutam (English Name)					
meṣaḥ [1] (Aries)	vṛṣabhaḥ [2] (Taurus)	mithunaḥ [3] (Gemini)	karkaḥ [4] (Cancer)	siṁhaḥ [5] (Leo)	kanyā [6] (Virgo)
tulā [7] (Libra)	vṛścikaḥ [8] (Scorpio)	dhanuḥ [9] (Saggitarius)	makaraḥ [10] (Capricorn)	kumbhaḥ [11] (Aquarius)	mīnaḥ [12] (Pisces)

Please note that the raasi are fixed regions in the sky (Sidereal and absolute), that are identified by the ‘clusters of stars’. Owing to the Vedic roots, these were identified and used the same way across all ancient cultures. The current western ‘Zodiac’, however, is Tropical based one that is pegged to the Month of the Solar Calendar – and hence has no relation (drifted over time) to the real constellations it was originally associated with.

maasa:

ṽmaa – to measure. Its the concept or idea of ‘quantifying’

ṽsa – to bind. Its the concept or idea of ‘putting together, such as in balance’. There are few variants to it, but they all kind of approximately convey the same idea.

From it came the derived words such as –

maatr and maatra = measurement of ‘things’. From it came English word ‘meter’ (through Greek), a measurement of length

samatva = equality, uniformity etc

maasa – From these dhaatu √maa and √sa came the word ‘maasa’ which means ‘uniformed measured’ and is roughly translated in English in the context of astronomy and astrology as ‘month’.

One complete Lunar Synodic revolution around Earth is called maasa and then there are 12 named maasa in a year (listed below). As mentioned earlier, the Moon completes 12 full Synodic revolutions (12 maasa) around Earth in a Synodic/Tropical Earth Year and that leaves us with few extra days (about 10) that goes into next maasa. The raasi divides the 360 degree sky into 12 equal units and acts as a ‘celestial scale’ which is absolute. So naturally, the ‘12 maasa year’ drift from the celestial scale raasi by about 10 days every year. The drift sums up to 1 full maasa in about 32 months. That’s where a new maasa is added with the same name as the succeeding one and with a prefix ‘adhika’, meaning ‘extra’. With such periodic addition, the 12 maasa are once again realigned with the raasi. In case of sayana calendar, which is Synodic/Tropical and accounts for precession, the progressive drift from the raasi is accounted for while adding the ‘adhika maasa’. In case of nirayana calendar, which is Sidereal and doesn’t account for precession, the ‘adhika maasa’ could be added to exactly align with the raasi.

The 12 maasa along with the reference raasi are:

maasa	raasi	maasa	raasi
chaitra [1]	mīnaḥ	ashvini [7]	kanyā
vaishakha [2]	meṣaḥ	kaartika [8]	tulā
jyeshtha [3]	vṛṣabhaḥ	margasheersha [9]	vṛścikaḥ
aashada [4]	mithunaḥ	poushya [10]	dhanuḥ
shraavana [5]	karkaḥ	maagha [11]	makaraḥ
baadhrapada [6]	simhaḥ	phalguna [12]	kumbhaḥ

graha:

√grah – to tie, grasp, fasten, captive etc. It’s the concept or idea of ‘holding’

From it came the derived words such as –

aagrah = eager, persistent, obstinate

nigrah = detain, constrain, restrict

graha – From this dhaatu √grah came the word ‘graha’ which means ‘one that holds, has an influence or pull’ and is roughly translated in English in the context of astronomy and astrology as ‘planet’.

With respect to Earth, it is not actually a ‘planet’ as we generally understand (inaccurately translated) today, but is really any celestial object that holds onto or has considerable influence on Earth. In that sense we have 7 ‘graha’. Then there are two celestial points of ‘reference’ for calculations called Rahu and Ketu – that are derived from the intersection line of Earth’s and Moon’s equatorial plane. The northern and southern points of intersecting line are respectively called Rahu and Ketu – and when moon is at those points and it is aligned with Sun, that causes Lunar and Solar Eclipse. Together they are referred to as the nava-graha, as in 9 graha.

The nava-graha (7 + 2 = 9) are:

graha name– samskrutam (English Name)		
surya, aditya (Sun)	chandra, soma (Moon)	Mangala (Mars)
budha (Mercury)	bruhaspathi, guru (Jupiter)	Shukra (Venus)
Shani (Saturn)	rahu (North intersecting point)	ketu (South intersecting point)

Shri Veda Vyasa wrote a stotram on the nava-graha and these are summarized in a verse –

*ādityāya ca somāya maṅgaldāya budhāya ca |
guru śukra śanibhyaśca rāhave ketave namaḥ ||*

From ‘Brihad Paraashara Hora Shastra’ (Chapter 3 Verse 10) –

*atha khetaa ravischandro mangalascha budhastatha |
guruh shukrah shanee raahuh ketuschaite yathakramam ||*

vaara:

The word vaara (or vaasarah) generally means a ‘week’ - derived from the Sanskrit dhaatu ‘vr’ which roughly means ‘to cover, encompass, surround’. Based on the Vedic roots the world over, in every ancient culture, a ‘week’ is generally accepted to have 7 days and they are based on the 7 identified graha and in the same order mentioned above.

The following table gives the names of the 7 vaara (days of the week) in the order they are defined in the vedic culture from the graha. It also has the names of the days in the week that are currently in use, along with their origins in other ancient cultures.

The 7 vaara (vaarah) are:

graha name (samskrutam)	vaara (vaasarah) name (samskrutam)	Name (English) L – Latin, G – Germanic, N – Norse R – Roman, O – Old English	Current Name (English)
ravi, surya, aditya	adi/ravi-vaasarah	Sun [L – Solis; G – Sunno]	Sun-day
chandra, soma	soma-vaasarah	Moon [L – Lunae; O – Monan]	Mon-day
mangala	mangala-vaasarah	Mars [R – Mars; L – Martis; G – Tiu]	Tues-day
budha	budha-vaasarah	Mercury [R – Mercury; L – Mercurii; G – Woden]	Wednes-day
bruhaspathi, guru	guru-vaasarah	Jupiter [R – Jupiter; L – Jovis; N – Thor]	Thurs-day
shukra	shukra-vaasarah	Venus [Roman – Venus; L – Veneris; N – Frigg]	Fri-day
shani	shani-vaasarah	Saturn [R – Saturn; L – Saturni]	Sat-day

nakshatra:

√ kshi – to possess, have power over

From it came the derived words such as –

kshetra = region, land, property

nakshatra – From this dhaatu *vkshi* came the word ‘na-kshatra’ which means ‘one that possesses no pull or power or falls outside the influencing region’ and is roughly translated in English in the context of astronomy and astrology as ‘stars’ - the ones, unlike ‘graha’, possess no significant gravitational power or influence on Earth.

The Synodic revolution of Moon around the Earth takes about 29.5 Earth Synodic days. The Sidereal revolution of Moon around the Earth takes little over 27 Earth Synodic days. The Synodic revolution of moon is well accounted for in tithi and raasi – with tithi dividing the 29.5 days into 30 equal units and raasi forming a 12 unit celestial scale for the 12 full revolutions in a Synodic year. The idea of nakshatra is similar to tithi but accounts for the Sidereal revolution of moon around Earth. Since tithi is Synodic revolution of moon around Earth, which as observed from distant star really revolves little more than 360 degrees, it can not be pegged to celestial scale – hence the 30 equal units were derived based on the 360 degree revolution as observed from Earth.

Initially, the little over 27 Earth Synodic days that the Sidereal revolution of Moon takes to go around Earth, were mapped out in the sky into 28 equal units pegged to prominent stars in each unit called nakshatra. Given that the sky is 360 degrees and perhaps since this lead to cumbersome division into

28 equal units ($360/28 = 12 \text{ deg } 51.429 \text{ arc minutes}$), this idea was later changed to dividing the sky into 27 equal units ($360/27 = 13 \text{ degrees } 20 \text{ arc minutes or } 800 \text{ arc minutes}$). Note – 1 degree = 60 arc minutes and 1 arc minute = 60 arc seconds. So nakshatra maps the 360 degree Sidereal revolution of moon around Earth (little over 27 Synodic Earth days) onto the stationary sky (a celestial scale) and divides them into 27 equal units. Each of these units are further divided into 4 pada (steps) – each being 3 degrees and 20 arc minutes or 200 arc minutes.

The 27 nakshatra are (Brihad Paraashara Hora Shastra (Chapter 3, Before Verse 7 – Notes from Santhanam):

nakshatra name – samskrutam		
ashwini [1]	maghā [10]	mula [19]
bharani [2]	pūrva phalgunī [11]	purva ashada [20]
krittika [3]	uttara phalgunī [12]	uttara ashada [21]
rohini [4]	hasta [13]	sravana [22]
mrigashīrsha [5]	chitra [14]	dhanishta [23]
ardra [6]	swāti [15]	shatabhisha [24]
punarvasu [7]	vshakha [16]	purva bhādrapadā [25]
pushya [8]	anuradha [17]	uttara bhādrapadā [26]
āshleshā [9]	jyeshtha [18]	revati [27]

yoga:

√ yuj – to join, put together. But is more precisely the concept of ‘synchronizing’ or ‘facilitating’ the elements of interest into ‘harmony’

From it came the derived words such as –

yojaka = a person preparing, organizing or arranging an event or act

yojana = the event or the act performed

yoga – From this dhaatu *vyuj* came the word ‘yoga’ which means ‘act of joining or putting together’ and has no equivalent in English in the context of astronomy and astrology.

As mentioned before, the Sidereal revolution of Moon around the Earth takes little over 27 Earth Synodic days. The nakshatra is similar to tithi – dividing the Moon’s revolution around earth into equal units. The tithi divides the Synodic revolution of Moon into 30 equal units of the apparent 360 degree revolution, whereas nakshatra divides the Sidereal revolution of Moon into 27 equal units of the real 360 degree revolution that is pegged to the sky (celestial scale). The 12 Synodic revolutions in a year are differentiated by pegging them to the celestial scale called raasi. The yoga, in that sense, is similar to raasi and is a way of differentiating the Sidereal revolutions of Moon around Earth. Since tithi is division of Synodic revolution, the raasi provides the differential celestial scale. However, the

nakshatra is division of Sidereal revolution that is already pegged to the celestial scale – so the yoga needs a slightly different method to differentiate the Sidereal revolutions.

We know that there are two Sidereal revolutions that are happening in this system – Moon around Earth and Earth around Sun. There are 27 yoga, just like there are 27 nakshatra. We start the first Moon's Sidereal revolution around Earth (using a reference nakshatra), by aligning the 1st of the 27 yoga with the 1st of the 27 nakshatra. Now we divide the Earth's 360 degree Sidereal revolution around Sun into 27 equal units ($360/27 = 13$ degrees 20 arc minutes or 800 arc minutes).

The duration of nakshatra (mentioned earlier) is Moon's angular revolution around Earth by 13 degrees 20 arc minutes. The duration of yoga is sum of Moon's angular revolution around Earth and Earth angular revolution around Sun being 13 degrees 20 arc minutes. This means that the duration of yoga is 1/27th fraction smaller than nakshatra. The concept is very similar to Vernier calipers.

The 27 yoga are:

yoga name – samskrutam		
viṣkambha [1]	gaṇḍa [10]	parigha [19]
prīti [2]	vṛddhi [11]	śiva [20]
āyusmān [3]	dhruva [12]	siddha [21]
saubhāgya [4]	vyāghatā [13]	sādhya [22]
śobhana [5]	harṣaṇa [14]	śubha [23]
atigaṇḍa [6]	vajra [15]	śukla [24]
sukarma [7]	siddhi [16]	brahma [25]
dhṛti [8]	vyatipāta [17]	māhendra [26]
śūla [9]	variya [18]	vaidhṛti [27]

karana:

√ kr – to do, make, perform, accomplish, cause, effect, prepare, undertake, manufacture, prepare, work at, elaborate, build form or construct one thing out of another.

From it came the derived words such as –

kaarya = to be made or done or practiced or performed

karana – From this dhaatu √kr came the word 'karana' which means 'doing, making, effecting, causing' and has no equivalent in English in the context of astronomy and astrology.

As already mentioned, a Synodic lunar month is divided into 30 equal parts called tithi and then the months are pegged against a celestial scale called raasi to account for the Synodic year. The Synodic

Earth days are further divided into hora (hour) and the year itself is grouped into 7 day units called vaasarah (weeks).

The karana is used to group the tithi into little units. The grouping is similar to vasarah for Synodic year. One karana is half a tithi. There are 30 tithi in a Synodic Lunar month. So there are 60 karana in a Synodic Lunar month. There are 11 types of karana. The last tithi is amavasya which consists of 2 karana. Of the 11 karana types, 2 karana are used only once for the amavasya tithi. One karana each are used for one before amavasya and one after amavasya (in the first tithi of the month – padyami). That leaves us with 7 types of karana that are used repetitively, spread across 28 tithi (27 full tithi and 2 half tithi) having 56 karana. These 7 types of karana are used 8 times repetitively (like vasarah), covering the 56 karana.

The 11 types of karana across the 30 tithi are:

karana	sukla/suddha paksha (waxing moon side) tithi name	karana	krishna/bahula paksha (waning moon side) tithi name
kimstughna [1]	pratipada/padyami [1]	bālava [3]	pratipada/padyami [16]
bava [2]		kaulava [4]	
bālava [3]	dwitiya/vidiya [2]	taitula [5]	dwitiya/vidiya [17]
kaulava [4]		garija [6]	
taitula [5]	tritiya/tadiya [3]	vaṇija [7]	tritiya/tadiya [18]
garija [6]		viṣṭi [8]	
vaṇija [7]	chaturthi/chaviti [4]	bava [2]	chaturthi/chaviti [19]
viṣṭi [8]		bālava [3]	
bava [2]	panchami [5]	kaulava [4]	panchami [20]
bālava [3]		taitula [5]	
kaulava [4]	shashti [6]	garija [6]	shashti [21]
taitula [5]		vaṇija [7]	
garija [6]	saptami [7]	viṣṭi [8]	saptami [22]
vaṇija [7]		bava [2]	
viṣṭi [8]	ashtami [8]	bālava [3]	ashtami [23]
bava [2]		kaulava [4]	
bālava [3]	navami [9]	taitula [5]	navami [24]
kaulava [4]		garija [6]	
taitula [5]	dashami [10]	vaṇija [7]	dashami [25]
garija [6]		viṣṭi [8]	
vaṇija [7]	ekadasi [11]	bava [2]	ekadasi [26]

viṣṭi [8]		bālava [3]	
bava [2]	dwadasi [12]	kaulava [4]	dwadasi [27]
bālava [3]		taitula [5]	
kaulava [4]	trayodasi [13]	garija [6]	trayodasi [28]
taitula [5]		vaṇija [7]	
garija [6]	chaturdasi [14]	viṣṭi [8]	chaturdasi [29]
vaṇija [7]		śakuna [9]	
viṣṭi [8]	pournami/poornima [15]	catuṣpāda [10]	amavasya [30]
bava [2]		nāga [11]	

Summary:

Real and Apparent motions – Earth spins about its axis, The axis wobbles (precession), Earth revolves around Sun, Moon spins about its axis (negligible), Moon revolves around Earth, Stars ‘revolve’ in the background.

Synodic – In short, it is a spin or revolution as ‘experienced’ from a ‘relative location’ such as Earth.

Sidereal – In short, it is a spin or revolution as seen from an ‘absolute location’ such as a distant star.

Precession – The Earth’s axis of spinning wobbles and completes the ‘wobble’ in about 26000 years.

ayana – It basically means precession.

saayana – The calendar that accounts for precession.

nirayana – The calendar that doesn’t account for precession.

Synodic/Tropical Solar Calendar – The year is based on the roughly 365 days Earth takes to revolve around Sun (adjusting additional time by adding days at suitable intervals). It accounts for the precession and divides the year into months and weeks based on the Earth Synodic days that come from its spinning. It is similar to the Gregorian Calendar.

Sidereal Solar Calendar – The year is based on one complete absolute revolution of Earth around Sun (not accounting for precession). Such calendars exist but has little practical value.

Synodic/Tropical Luni-Solar Calendar – The year is based on the roughly 365 days Earth takes to revolve around Sun (adjusting additional time by adding months at suitable intervals). It accounts for the precession and divides the year into fixed size months based on the Synodic Lunar revolution around Earth. It has weeks (vaasarah) based on the Earth Synodic days that come from its spinning and tithi, which is dividing the Lunar revolution into 30 equal units. This is the most commonly used calendar.

panchanga – It literally translates to ‘five limbs’ (pancha + anga). It is a Sidereal-Synodic based complex Calendar. The five ‘limbs’ are tithi, vaara, nakshatra, yoga and kaarana – though there are several other concepts that are accounted for such as graha, vaasarah, maasa, raasi etc.

tithi – It is the 360 degree Synodic revolution of Moon around Earth divided into 30 equal units.

raasi – It is a fixed ‘celestial scale’ of 12 units in 360 degrees (number of full Synodic Lunar revolutions around Earth in a Synodic Year), through which our Earth is navigating.

maasa – This is the fixed size ‘month’ based on the 12 full Synodic Lunar revolutions around Earth in a Synodic Year

graha – These are 9 in number. The 7 celestial objects that have tangible gravitational influence Earth along with 2 celestial reference points.

vaara – It is a ‘week’ that is 7 Earth Synodic days long and is based on the 7 identified graha in a specific order.

hora – It is an ‘hour’ and there are 24 of those in a Earth Synodic day that is used for the days in the vaasarah.

nakshatra – It is a fixed ‘celestial scale’ of 27 units ($360/27 = 13$ degrees 20 arc minutes or 800 arc minutes) – which is pegged to the Moon to complete one absolute 360 degrees Sidereal revolution around Earth. The number comes from roughly 27 Synodic Days Earth takes for the Moon to complete such revolution.

yoga – It is a fixed scale of 27 units (13 degrees 20 arc minutes each) that is imposed over 27 nakshatra. However, this takes into account both the Sidereal revolution of Moon around Earth and Sidereal revolution of Earth around Sun. A yoga changes when the summation of both angular revolutions (of 360 degrees each) comes to 13 degrees and 20 arc minutes. This makes yoga shorter than nakshatra by about 1/27th fraction. The concept is very similar to Vernier calipers.

karana – A karana is half of tithi and so there are 60 karana in a 30 tithi Synodic Lunar month. There are 11 types of karana. It functions similar to vaasarah. The 2 karana on amavasya tithi along with one karana each before and after amavasya, a total of 4 karana, are one type each. The remaining 7 types of karana are repeated sequentially like for vaasarah, a total of 8 times ($56/7$) to cover for the rest of the month.

Notes:

Festivals – Most of the Hindu festivals such as vijayadashami, ugadi, diwali, holi etc follow either panchanga calendar or the simpler variant Synodic Luni-Solar calendar – celebrated on specific days of the lunar months. So in every sense they fall on the ‘same day’ if that Hindu calendar is followed. The ‘day’ obviously and naturally ‘varies’ and falls on ‘different days’ only on a Solar calendar (such as on Gregorian calendar).

sankranthi – The word sankranthi comes from ‘sankramana’, meaning to ‘cross over or transit’. The term is referred to the ‘crossing over’ of Earth from one raasi to another. So there are 12 sankranthi in

a year. The 'makar sankranti', which is the most celebrated one, refers to the one where Earth transits into the makar raasi. This is a Sidereal event – as in, irrespective of Earth's precession (Synodic event that drifts seasons across celestial scale), the celebration will always be pegged to the fixed celestial scale (raasi) and occurs when Earth transits into the makar raasi. So on any Synodic/Tropical calendar (such as Gregorian Solar calendar), which accounts for precession, the date will always drift – by about 20 minutes per year (that is 1 day every 72 years).

Reference Year – All the Solar calendars necessarily require a reference year to start the count, because every year is identical in nature and can only be differentiated by counting from a reference year. The scales in panchanga are much larger running into thousands of years with unique timestamp, owing to the multiple data points that are used to peg a specific date. The ones mentioned above in the panchanga section are only a few data points – there are many more such as going from the fractions of seconds to cycles of years.

Sonti Mohan Krishna (2020)